

Foundations of Data Science

Fall 2026

Prof. Gilles Louppe
g.louppe@uliege.be

Course hub

Instructions and materials are all available on GitHub
github.com/glouppe/dats0001-foundations-of-data-science:

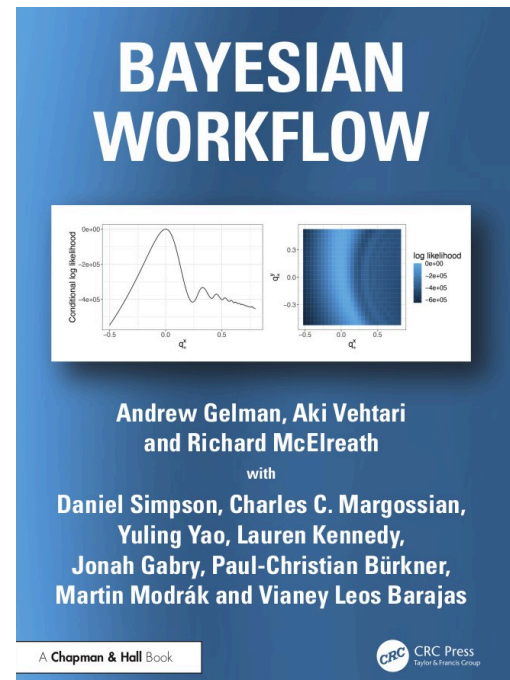
- Schedule
- Slides and notebooks
- Project

Official course description: [engagements pédagogiques](#).

Main reference

Gelman, Vehtari, McElreath et al., [Bayesian Workflow](#), Chapman & Hall/CRC, 2026.

- Electronic edition freely available for non-commercial use.
- Relevant chapters are listed in the readings of each lecture.
- Case studies are in R and Stan (we use Python for the course).



Lectures

- Mondays, 1:30 PM
- Come with your laptop!

Case study

- Practice your data science skills through an end-to-end analysis.
- Presentation/defense of your analysis and results.

Exam

Oral exam on theory covered in class.

Evaluation

- Case study (40%)
- Oral exam (60%)

A minimum of 8/20 is required for each component. Below that, the lowest of the two grades becomes the final grade.

Use of AI tools

AI assistants are **allowed**, without restriction.

The rule is simple: everything you submit, you must be able to explain. Submitting an analysis you do not understand, because an assistant produced it, will be strongly penalized.